

COLDPOD · REQUIREMENTS

ColdPod requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.9	2026-10-02	Amish Chadha	Status against CPD-CAL-001 v0.7: battery shipping switch and R-134a charge added; R12 5.74 kg (0.01 kg margin); R15 \$313, \$3 over the value-engineering target
0.8	2026-10-02	Amish Chadha	Decisions of 2026-10-02: R8 restated as 9 h, R12 relaxed to 5.75 kg for the prototype, R14 removal method for the prototype; R8 and R12 now met on paper
0.7	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.6	2026-10-01	Amish Chadha	Status values from CPD-CAL-001 v0.4 for the constructable design (CPD-DDR-003, Draft); no target changed
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish; R15 target \$310, status met against CPD-CAL-001 v0.3
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). R2 and R7 targets restated for the liner cut-out and

			buck-boost driver; status from CPD-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	Record Amish's decisions (CPD-DDR-001); R6 redefined to 16 h, R12 relaxed to 5.5 kg, R15 set to \$300; status from CPD-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

ColdPod requirements

These are the requirements for the concept, checked by calculation at TRL 3 in CPD-CAL-001 v0.7 against the constructable design of CPD-DDR-003, which Amish accepted on 2026-10-02. Amish decided the TRL 2 review items on 2026-09-25 (CPD-DDR-001): R6 is redefined from 24 h to 16 h at 43 °C (D5), R12 is relaxed from 5.0 kg to 5.5 kg (D6), R15 follows the \$300 value-engineering target of D10 and the R10 alarm defaults are adopted (D9). He then accepted the TRL 3 recommendations (CPD-DDR-002): R2 now names the two hardware cut-outs in series and R7 names the buck-boost driver that serves the full 10 to 15 V vehicle range; no target was relaxed. The target is now \$310 (CPD-DDR-002 N3), a hypothetical control target and not a limit, and R15 follows it. No target changed in v0.6; the values follow the constructable design. Against CPD-CAL-001 v0.4, ten requirements were met, three at risk, two **not met** by small margins (R8 and R12) and two not verifiable at TRL 3 (R10 and R16). On 2026-10-02 Amish relaxed R12 to 5.75 kg for the prototype and restated R8 as 9 h (CPD-DEC-001), so twelve were met on paper; after the battery shipping switch was added (BOM line 18, \$5) the cost is \$313 and R15 is \$3 over its target, so eleven are met, three at risk, one not met (R15) and two not verifiable; R14's removal method was decided for the prototype. Before DDR-002 the count was seven met, five at risk and three not met (R2, R8, R12).

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (CPD-CAL-001)
R1	Keep the payload in the safe range	2 to 8 °C at a buffered probe anywhere in the payload space; control setpoint 5 °C	Heat-transfer calculation; later lab test in a climate chamber	Met on paper: liner held at 2 to 5 °C; the lid pack stays frozen near 2.8 °C on its cold plate in powered hold at 43 °C (was at risk)
R2	Never freeze the payload	No payload-contact surface below 1 °C in normal operation or with a stuck-on Peltier driver; two hardware cut-outs in series in the Peltier supply, on the cold block (–5 °C) and on the liner (3 °C), independent of firmware (CPD-DDR-002)	Calculation of the cold path; fault analysis	Met on paper: with a stuck-on driver the liner settles near 2.0 °C after the liner cut-out opens (was not met, about –4 °C)
R3	Carry a useful load	1.0 L or more of usable payload; holds 20 or more insulin pens up to 170 mm long	Massing model	Met: 1.37 L, 24 pens, up to 172 mm
R4	Hold with no power at all	12 h or more at a constant +43 °C, starting with the PCM frozen	Heat-leak and PCM calculation	At risk: 13.3 h; 10.2 h with 30 % more heat leak
R5	Hold off-grid on a full charge	24 h or more at a constant +32 °C on the internal battery, then the PCM	Energy budget	At risk: 27.3 h; 21.4 h with 30 % more heat leak
R6	Hold off-grid on a full charge in extreme heat	16 h or more at a constant +43 °C on the internal battery, then the PCM; longer hot trips run from external power under R7 (redefined from 24 h, CPD-DDR-001 D5)	Energy budget	At risk: 16.7 h; 12.6 h with 30 % more heat leak
R7	Hold indefinitely on external power	Keeps 2 to 8 °C up to +43 °C from 12 V DC (10 to 15 V, through a buck-boost Peltier driver, CPD-DDR-002) or USB-C PD (20 V, 45 W or more)	Peltier and heat sink calculation	Met on paper: 18.8 W input, well inside 45 W; the buck-boost stage covers a 10 V input; the lid pack stays frozen (was at risk)
R8	Recharge the cold quickly	Refreeze a fully melted PCM in 9 h or less at 25 °C ambient, box empty, while charging the battery (restated from 8 h, CPD-DEC-001, 2026-10-02; still an overnight refreeze)	PCM and Peltier calculation	Met: 8.0 h (8.03 h) for all the PCM (jacket 6.8 h, lid pack 8.0 h); 8.4 h before the constructable design, which holds slightly less PCM
R9	Record the temperature	Payload probe accuracy ± 0.5 °C; log every 1 min; 60 days or more on board; export as CSV over Bluetooth Low Energy or USB	Datasheet and storage calculation; later calibration	Met: 1.38 MB of 2.10 MB; accuracy by sensor selection
R10	Raise alarms early	Local sound, light and display alarm, plus a phone notification when paired: warn after 10 min outside 2 to 8 °C; alarm at once at 0 °C or lower on the payload probe; early freeze warning at 1 °C on the liner probe; low battery, sensor fault and lid open for more than 2 min (defaults decided, CPD-DDR-001 D9; adjustable per product)	Design review of firmware sketch	Not verifiable at TRL 3: design intent only, no firmware sketch
R11			Power budget	Met: about 96 days

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (CPD-CAL-001)
	Keep logging when the cooling stops	Logger runs 14 days or more after the Peltier is shut off for low battery		
R12	Be light enough to carry all day	5.75 kg (12.7 lb) or less empty for the prototype (relaxed from 5.0 kg, CPD-DDR-001 D6, and from 5.5 kg, CPD-DDR-003 A2, 2026-10-02)	Mass estimate, weighing at TRL 4	Met on paper: 5.74 kg, 0.01 kg margin (0.05 kg before the battery shipping switch and the R-134a charge were added) (5.53 kg before the fixings and parts added to make the design buildable, CPD-DDR-003); no base bumper until the prototype is weighed
R13	Be compact	Fits in 400 x 250 x 250 mm including handle	Massing model	Met: 366 x 220 x 207 mm
R14	Travel by air	Battery 100 Wh or less, removable or with a shipping switch	Battery specification	Met for the prototype: 76.8 Wh; the pack comes out with the battery bay (four screws), counted as removable for the prototype (CPD-DDR-003 A3, 2026-10-02). Before any field or air travel use: a shipping switch and the carrier's lithium battery rules
R15	Be affordable and buildable	Estimated parts cost within the \$310 value-engineering target (hypothetical control target, CPD-DDR-002 N3); no custom PCB for the first build	Priced BOM	Over the value-engineering target by \$3: \$313 against \$310, after the \$5 battery shipping switch (BOM line 18) added on 2026-10-02
R16	Survive field use	Splash resistant (IP54 target for electronics bays); survives a 0.5 m drop onto a hard floor while loaded	Design review; later test	Not verifiable at TRL 3
R17	Be clearly labelled as a prototype	"Research prototype, not a medical device" on the box, the display start screen and every exported log	Design review	Met by design

■ Assumptions

- Ambient cases follow the WHO PQS test point of +43 °C for the hot zone ([WHO PQS E004/VC01](#)), +32 °C for a typical hot day and 25 °C for a clinic.
- The 2 to 8 °C window and the no-freezing rule come from insulin and vaccine storage guidance ([US FDA](#); [CDC Pink Book, chapter 5](#)).
- Logging and alarm targets are stricter than a WHO 30-day recorder (−0.5 °C for 60 min, +8 °C for 10 h; [TechNet-21](#)) because a carrier in transit can move out of range much faster than a refrigerator.
- R6 matches the WHO long-range carrier cold life only loosely: a WHO carrier is tested passive for 30 h at 43 °C. ColdPod does not claim WHO equivalence.
- The alarm thresholds in R10 were decided by Amish on 2026-09-25 (CPD-DDR-001 D9) and will be adjustable per product.

- Hold-time status uses a $\pm 30\%$ band on the heat leak: met if the target holds at $+30\%$, at risk if it holds only at the nominal estimate (CPD-CAL-001).
- R15 counts ColdPod parts only; the payload is not costed.